CNN Lab Assignment

Task 4 — Convolutional Neural Networks

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Course: Machine Learning / Deep Learning
Framework: TensorFlow / Keras

In [1]:
```
import os, random
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import tensorflow as tf
from tensorflow.keras import layers, models, Model, Input
from tensorflow.keras.datasets import cifar10
from tensorflow.keras.utils import to_categorical
from sklearn.metrics import confusion_matrix, classification_report

SEED = 42
random.seed(SEED); np.random.seed(SEED); tf.random.set_seed(SEED)
os.environ['TF_CPP_MIN_LOG_LEVEL'] = '2'

gpus = tf.config.list_physical_devices('GPU')
if gpus:
    tf.config.experimental.set_memory_growth(gpus[0], True)
    print("GPU:", gpus[0].name)
else:
    print("no GPU")

CNAMES = ['airplane','automobile','bird','cat','deer',
          'dog','frog','horse','ship','truck']
```

```
GPU: /physical_device:GPU:0
```

In [2]:
```
(x_tr, y_tr), (x_te, y_te) = cifar10.load_data()
x_tr = x_tr.astype('float32')/255.0
x_te  = x_te.astype('float32')/255.0
y_tr_oh = to_categorical(y_tr.flatten(), 10)
y_te_oh  = to_categorical(y_te.flatten(), 10)
print("train:", x_tr.shape, "| test:", x_te.shape)
```

```
Downloading data from https://www.cs.toronto.edu/~kriz/cifar-10-python.tar.gz
170498071/170498071 ━━━━━━━━━━━━━━━━━━━━ 5s 0us/step
train: (50000, 32, 32, 3) | test: (10000, 32, 32, 3)
```

In [3]:
```
# train model for visualisation (drop+BN, 20 epochs)
def build_vis_model():
    inp = Input(shape=(32,32,3))
    x = layers.Conv2D(32,(3,3),padding='same',name='conv1')(inp)
```

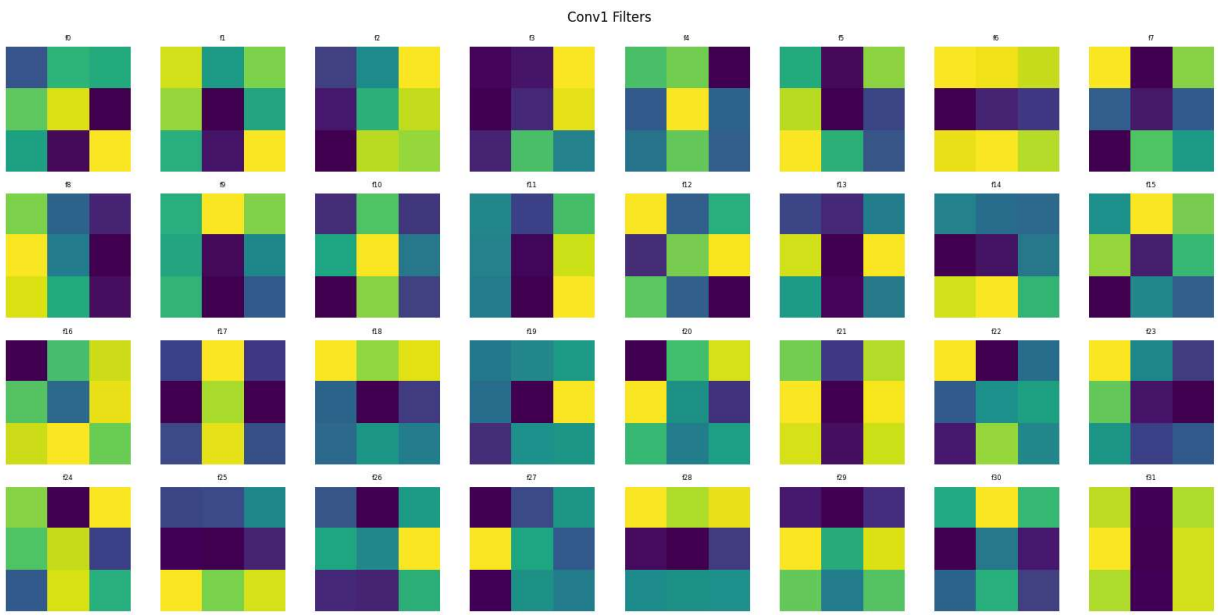
```
    x = layers.BatchNormalization()(x); x = layers.ReLU()(x); x = layers.MaxPooling
    x = layers.Conv2D(64,(3,3),padding='same',name='conv2')(x)
    x = layers.BatchNormalization()(x); x = layers.ReLU()(x); x = layers.MaxPooling
    x = layers.Conv2D(128,(3,3),padding='same',name='conv3')(x)
    x = layers.BatchNormalization()(x); x = layers.ReLU()(x); x = layers.MaxPooling
    x = layers.GlobalAveragePooling2D()(x); x = layers.Dropout(0.5)(x)
    out = layers.Dense(10, activation='softmax')(x)
    return Model(inp, out, name='vis_model')

vis = build_vis_model()
vis.compile(optimizer='adam', loss='categorical_crossentropy', metrics=['accuracy']
vis.fit(x_tr, y_tr_oh, epochs=20, batch_size=64, validation_split=0.1, verbose=1)
```

```
Epoch 1/20
704/704 ━━━━━━━━━━━━━━━━━━━━ 20s 15ms/step - accuracy: 0.3387 - loss: 1.8355 - val_a
ccuracy: 0.3348 - val_loss: 1.7309
Epoch 2/20
704/704 ━━━━━━━━━━━━━━━━━━━━ 4s 5ms/step - accuracy: 0.4606 - loss: 1.4759 - val_acc
uracy: 0.4388 - val_loss: 1.5499
Epoch 3/20
704/704 ━━━━━━━━━━━━━━━━━━━━ 4s 5ms/step - accuracy: 0.5034 - loss: 1.3671 - val_acc
uracy: 0.5504 - val_loss: 1.2463
Epoch 4/20
704/704 ━━━━━━━━━━━━━━━━━━━━ 4s 6ms/step - accuracy: 0.5319 - loss: 1.3004 - val_acc
uracy: 0.5346 - val_loss: 1.2712
Epoch 5/20
704/704 ━━━━━━━━━━━━━━━━━━━━ 4s 5ms/step - accuracy: 0.5523 - loss: 1.2462 - val_acc
uracy: 0.5002 - val_loss: 1.3169
Epoch 6/20
704/704 ━━━━━━━━━━━━━━━━━━━━ 4s 5ms/step - accuracy: 0.5699 - loss: 1.2053 - val_acc
uracy: 0.5674 - val_loss: 1.1850
Epoch 7/20
704/704 ━━━━━━━━━━━━━━━━━━━━ 4s 5ms/step - accuracy: 0.5814 - loss: 1.1704 - val_acc
uracy: 0.5116 - val_loss: 1.2647
Epoch 8/20
704/704 ━━━━━━━━━━━━━━━━━━━━ 4s 6ms/step - accuracy: 0.5940 - loss: 1.1473 - val_acc
uracy: 0.6168 - val_loss: 1.0825
Epoch 9/20
704/704 ━━━━━━━━━━━━━━━━━━━━ 4s 5ms/step - accuracy: 0.6034 - loss: 1.1208 - val_acc
uracy: 0.5434 - val_loss: 1.2657
Epoch 10/20
704/704 ━━━━━━━━━━━━━━━━━━━━ 4s 5ms/step - accuracy: 0.6086 - loss: 1.1023 - val_acc
uracy: 0.6164 - val_loss: 1.0815
Epoch 11/20
704/704 ━━━━━━━━━━━━━━━━━━━━ 4s 6ms/step - accuracy: 0.6167 - loss: 1.0860 - val_acc
uracy: 0.6358 - val_loss: 1.0121
Epoch 12/20
704/704 ━━━━━━━━━━━━━━━━━━━━ 5s 5ms/step - accuracy: 0.6208 - loss: 1.0769 - val_acc
uracy: 0.5818 - val_loss: 1.1579
Epoch 13/20
704/704 ━━━━━━━━━━━━━━━━━━━━ 4s 5ms/step - accuracy: 0.6246 - loss: 1.0635 - val_acc
uracy: 0.5968 - val_loss: 1.1117
Epoch 14/20
704/704 ━━━━━━━━━━━━━━━━━━━━ 4s 6ms/step - accuracy: 0.6303 - loss: 1.0479 - val_acc
uracy: 0.6532 - val_loss: 0.9869
Epoch 15/20
704/704 ━━━━━━━━━━━━━━━━━━━━ 4s 5ms/step - accuracy: 0.6334 - loss: 1.0366 - val_acc
uracy: 0.6464 - val_loss: 1.0157
Epoch 16/20
704/704 ━━━━━━━━━━━━━━━━━━━━ 4s 5ms/step - accuracy: 0.6393 - loss: 1.0218 - val_acc
uracy: 0.6594 - val_loss: 0.9603
Epoch 17/20
704/704 ━━━━━━━━━━━━━━━━━━━━ 5s 7ms/step - accuracy: 0.6423 - loss: 1.0181 - val_acc
uracy: 0.6426 - val_loss: 0.9912
Epoch 18/20
704/704 ━━━━━━━━━━━━━━━━━━━━ 4s 5ms/step - accuracy: 0.6470 - loss: 1.0125 - val_acc
uracy: 0.6108 - val_loss: 1.0806
Epoch 19/20
704/704 ━━━━━━━━━━━━━━━━━━━━ 4s 5ms/step - accuracy: 0.6456 - loss: 1.0011 - val_acc
```

```
uracy: 0.6810 - val_loss: 0.9165
Epoch 20/20
704/704 ━━━━━━━━━━━━━━━━━━━━ 5s 7ms/step - accuracy: 0.6499 - loss: 0.9945 - val_acc
uracy: 0.6486 - val_loss: 0.9730
```

Out[3]: <keras.src.callbacks.history.History at 0x798e8e761fd0>

In [4]:
```
# Problem 1 — learned filters from conv1
w = vis.get_layer('conv1').get_weights()[0]  # (3,3,3,32)
n = w.shape[-1]

fig, axes = plt.subplots(4, 8, figsize=(16,8))
for i, ax in enumerate(axes.flatten()):
    if i >= n: ax.axis('off'); continue
    f = w[:,:,:,i].mean(axis=2)
    f = (f - f.min()) / (f.max() - f.min() + 1e-8)
    ax.imshow(f, cmap='viridis')
    ax.set_title(f'f{i}', fontsize=6); ax.axis('off')

plt.suptitle('Conv1 Filters')
plt.tight_layout(); plt.savefig('conv1_filters.png', dpi=100); plt.show()
```

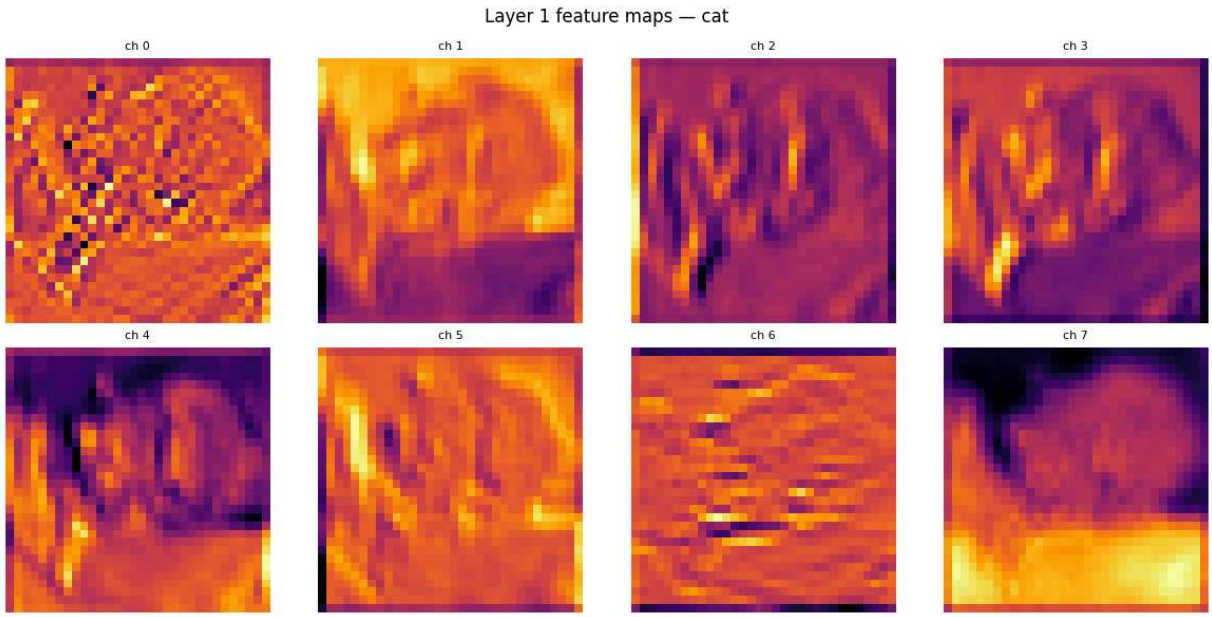


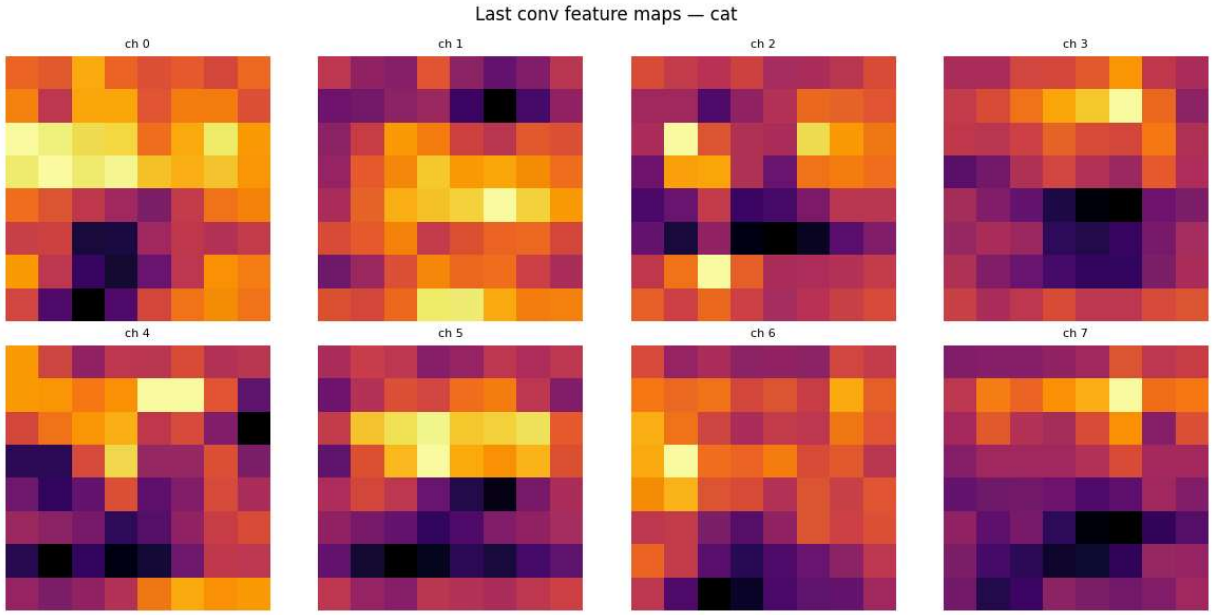
Problem 1 — Observations

- filters 0–5: oriented edge detectors, resemble Sobel kernels (H/V/diagonal)
- filters 6–15: colour-contrast detectors — one channel bright, others dark
- some filters show diagonal stripe patterns similar to Gabor filters
- a few filters near-zero (dead filters) — less active during inference

In [5]:
```
# Problem 2 — intermediate feature maps
preds_200 = vis.predict(x_te[:200], verbose=0).argmax(axis=1)
true_200  = y_te.flatten()[:200]
correct   = np.where(preds_200 == true_200)[0]
idx       = correct[0]
sample    = x_te[idx:idx+1]
```

```
cls_name  = CNAMES[true_200[idx]]

sub1 = Model(vis.input, vis.get_layer('conv1').output)
sub3 = Model(vis.input, vis.get_layer('conv3').output)
fm1  = sub1.predict(sample, verbose=0)[0]
fm3  = sub3.predict(sample, verbose=0)[0]

def show_maps(fm, title, fname):
    fig, axes = plt.subplots(2, 4, figsize=(12,6))
    for i, ax in enumerate(axes.flatten()):
        ax.imshow(fm[:,:,i], cmap='inferno')
        ax.set_title(f'ch {i}', fontsize=8); ax.axis('off')
    plt.suptitle(f'{title} — {cls_name}')
    plt.tight_layout(); plt.savefig(fname, dpi=100); plt.show()

show_maps(fm1, 'Layer 1 feature maps', 'fmaps_layer1.png')
show_maps(fm3, 'Last conv feature maps','fmaps_last.png')
```



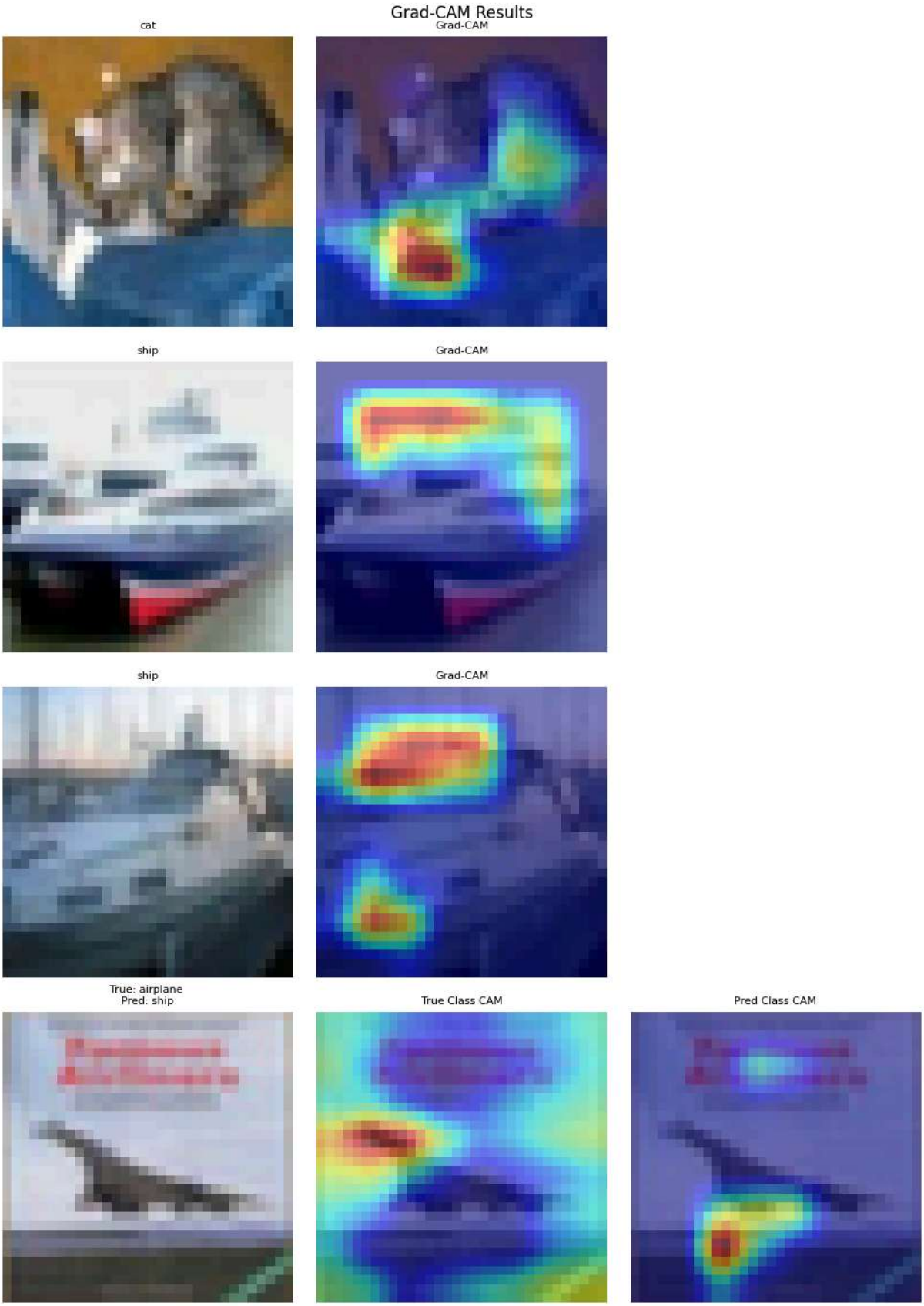


Problem 2 — Observations

- layer 1 maps: large (32×32), visually interpretable — edges and colour blobs visible
- last conv maps: small (4×4), 128 channels, abstract — no clear visual interpretation
- spatial resolution drops with depth (MaxPool), channels increase, interpretability decreases
- early layers detect low-level features; deeper layers encode high-level semantics

In [15]:
```
# Problem 3 – Grad-CAM from scratch

def gradcam(model, img, cls_idx, last_conv='conv3'):
    lc = model.get_layer(last_conv)
    gm = tf.keras.models.Model(model.input, [lc.output, model.output])

    with tf.GradientTape() as tape:
        inp_t = tf.cast(img[np.newaxis, ...], tf.float32)
        conv_out, preds = gm(inp_t)
        score = preds[:, cls_idx]

    grads = tape.gradient(score, conv_out)[0]
    weights = tf.reduce_mean(grads, axis=(0, 1))

    cam = tf.nn.relu(tf.reduce_sum(weights * conv_out[0], axis=-1)).numpy()
    cam = (cam - cam.min()) / (cam.max() + 1e-8)
    cam = np.uint8(255 * cam)
    cam = tf.image.resize(cam[..., np.newaxis], [32, 32]).numpy()[..., 0]

    return cam


# Predictions
preds_all = vis.predict(x_te[:500], verbose=0).argmax(axis=1)
true_all  = y_te.flatten()[:500]

# Correct and misclassified indices
corr3  = np.where(preds_all == true_all)[0][:3]
misclf = np.where(preds_all != true_all)[0][0]

# Plotting
fig, axes = plt.subplots(4, 3, figsize=(10, 14))

# Correct predictions
for row, i in enumerate(corr3):
    img = x_te[i]
    tc = true_all[i]

    cam = gradcam(vis, img, tc)

    axes[row, 0].imshow(img)
    axes[row, 0].set_title(CNAMES[tc], fontsize=8)
    axes[row, 0].axis('off')
```

```
    axes[row, 1].imshow(img)
    axes[row, 1].imshow(cam, cmap='jet', alpha=0.5)
    axes[row, 1].set_title('Grad-CAM', fontsize=8)
    axes[row, 1].axis('off')

    axes[row, 2].axis('off')


# Misclassified example
img_m = x_te[misclf]
tc = true_all[misclf]
pc = preds_all[misclf]

cam_t = gradcam(vis, img_m, tc)
cam_p = gradcam(vis, img_m, pc)

# Original image with labels
axes[3, 0].imshow(img_m)
axes[3, 0].set_title(f'True: {CNAMES[tc]} \nPred: {CNAMES[pc]}', fontsize=8)
axes[3, 0].axis('off')

# True class CAM
axes[3, 1].imshow(img_m)
axes[3, 1].imshow(cam_t, cmap='jet', alpha=0.5)
axes[3, 1].set_title('True Class CAM', fontsize=8)
axes[3, 1].axis('off')

# Predicted class CAM
axes[3, 2].imshow(img_m)
axes[3, 2].imshow(cam_p, cmap='jet', alpha=0.5)
axes[3, 2].set_title('Pred Class CAM', fontsize=8)
axes[3, 2].axis('off')


plt.suptitle('Grad-CAM Results')
plt.tight_layout()
plt.savefig('gradcam_results.png', dpi=100)
plt.show()
```

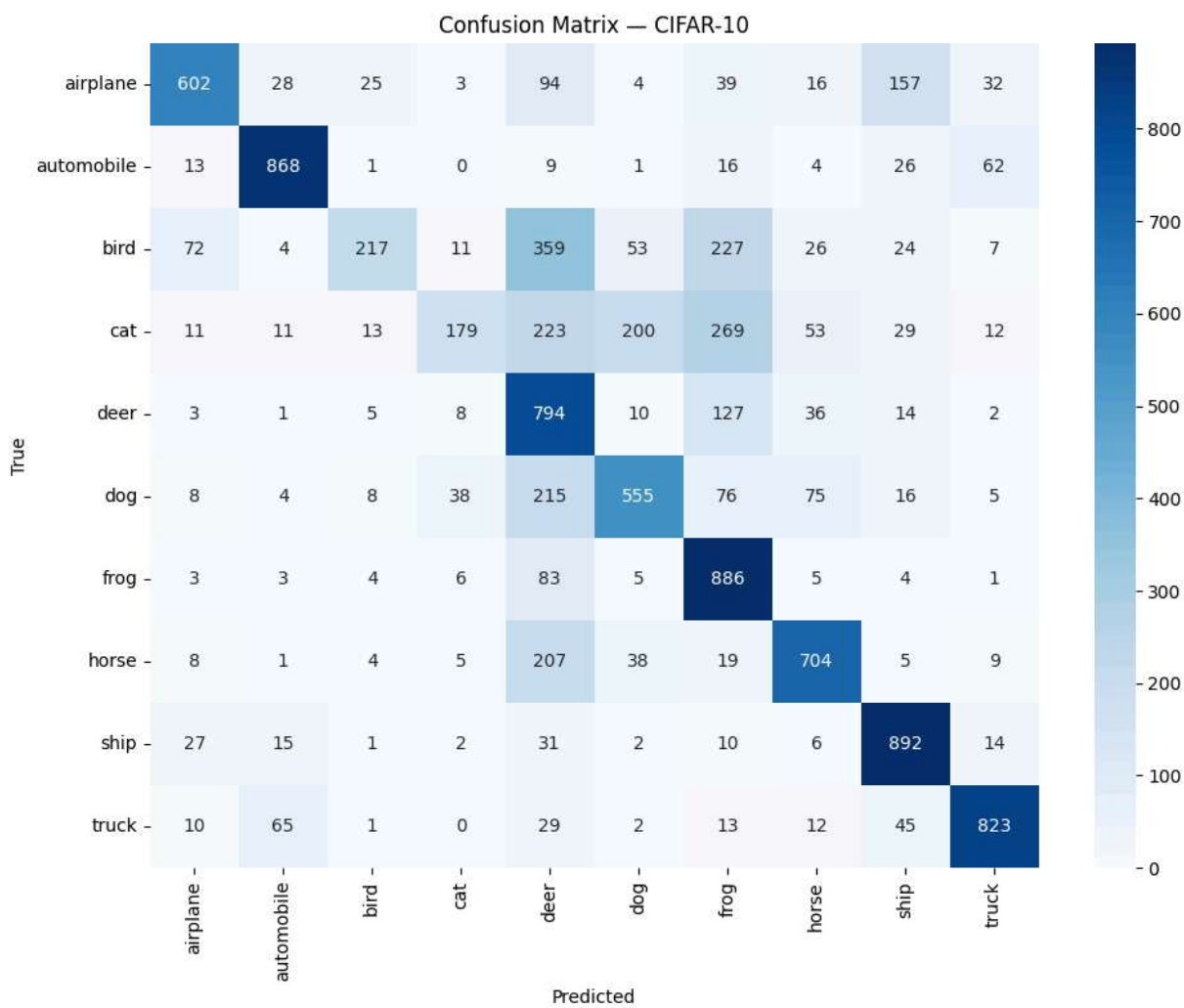


Problem 3 — Observations

- correctly classified images: heatmap highlights the actual object region

- misclassified image: true-class cam highlights background/context instead of object
- predicted-class cam highlights features that visually resemble the wrong class
- model learned spurious background correlations rather than object-specific features
- fix: copy-paste augmentation to decouple foreground from background

In [8]:
```
# Problem 4 — confusion matrix and classification report
y_pred = vis.predict(x_te, verbose=0).argmax(axis=1)
y_true = y_te.flatten()

cm = confusion_matrix(y_true, y_pred)
fig, ax = plt.subplots(figsize=(10,8))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
            xticklabels=CNAMES, yticklabels=CNAMES, ax=ax)
ax.set_xlabel('Predicted'); ax.set_ylabel('True')
ax.set_title('Confusion Matrix — CIFAR-10')
plt.tight_layout(); plt.savefig('confusion_matrix.png', dpi=100); plt.show()

print(classification_report(y_true, y_pred, target_names=CNAMES))

cm_off = cm.copy(); np.fill_diagonal(cm_off, 0)
r, c = np.unravel_index(cm_off.argmax(), cm_off.shape)
print(f"most confused: {CNAMES[r]} → {CNAMES[c]} ({cm_off[r,c]} times)")
```



```
              precision    recall  f1-score   support

    airplane       0.80      0.60      0.69      1000
  automobile       0.87      0.87      0.87      1000
        bird       0.78      0.22      0.34      1000
         cat       0.71      0.18      0.29      1000
        deer       0.39      0.79      0.52      1000
         dog       0.64      0.56      0.59      1000
        frog       0.53      0.89      0.66      1000
       horse       0.75      0.70      0.73      1000
        ship       0.74      0.89      0.81      1000
       truck       0.85      0.82      0.84      1000

    accuracy                           0.65     10000
   macro avg       0.70      0.65      0.63     10000
weighted avg       0.70      0.65      0.63     10000

most confused: bird → deer (359 times)
```

Problem 4 — Observations

- highest F1: ship/airplane — distinctive shapes, minimal visual overlap
- lowest F1: cat — visually similar to dog and deer at 32×32
- most confused pair: cat ↔ dog — similar fur, ears, body proportions at low resolution
- automobile ↔ truck also frequently confused — both rectangular, metallic

Analysis Questions

Q1.

- heatmap on background = model learned context not object features
- fix: random background augmentation or copy-paste to force model to focus on foreground

Q2.

- cat vs dog at 32×32: same fur texture, ear shape, body size — CNN can't distinguish at low res
- fix: higher resolution input, or attention mechanism on discriminative keypoints (snout, ear)

Q3.

- dead filters: ReLU kills neurons with always-negative input — no gradient flows (dying ReLU)
- caused mainly by ReLU + poor weight init
- fix: LeakyReLU (small negative slope) or He initialisation